blob will typically undulate until it settles into the shape of a sphere, but it needs sufficient time to harden before hitting the bottom. On orbiting space stations, where everything is weightless, you gently squirt out precise quantities of molten metal and you have all the time you need—the beads just float there while they cool, until they harden as perfect spheres, with surface tension doing all the work for you.

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For large cosmic objects, energy and gravity conspire to turn objects into spheres. Gravity is the force that serves to collapse matter in all directions, but gravity does not always win—chemical bonds of solid objects are strong. The Himalayas grew against the force of Earth's gravity because of the resilience of crustal rock. But before you get excited about Earth's mighty mountains, you should know that the spread in height from the deepest undersea trenches to the tallest mountains is about a dozen miles, yet Earth's diameter is nearly eight thousand miles. So, contrary to what it looks like to teeny humans crawling on its surface, Earth, as a cosmic object, is remarkably smooth. If you had a super-duper, jumbo-gigantic finger, and you dragged it across Earth's surface (oceans and all), Earth would feel as smooth as a cue ball. Expensive globes that portray raised portions of Earth's landmasses to indicate mountain ranges are gross exaggerations of reality. This is why, in spite of Earth's mountains and valleys, as well as being slightly flattened from pole to pole, when viewed from space, Earth is indistinguishable from a perfect sphere.

Earth's mountains are also puny when compared with some other mountains in the solar system. The largest on Mars, Olympus Mons, is 65,000 feet tall and nearly 300 miles wide at its base. It makes Alaska's Mount McKinley look like a molehill. The cosmic mountain-building recipe is simple: the weaker the gravity on the surface of an object, the higher its mountains can reach. Mount Everest is about as tall as a mountain on Earth can grow before the lower rock layers succumb to their own plasticity under the mountain's weight.

If a solid object has a low enough surface gravity, the chemical bonds in its rocks will resist the force of their own weight. When this happens, almost any shape is possible. Two famous celestial non-spheres are Phobos and Deimos, the Idaho potato–shaped moons of Mars. On thirteen-mile-long Phobos, the bigger of the two moons, a 150-pound person would weigh a mere four ounces.